Odd and dyadic local geometries share reconstruction, not integral weights

coordinate	odd prime p	dyadic prime 2
torsion address	$j \bmod (p-1)$	sign and parity
principal units	$1+p\mathbb{Z}_p$	$1+4\mathbb{Z}_2$
base Mahler gain	one p adic digit	three binary digits
Euler source	remove one nonunit mode	already unit normalized
Euler kernel depth	$2m+v_p(Z_{2m+1})$	$4m-1$

Character phase, principal unit depth, source normalization, and kernel valuation must be tracked separately.